UNIVERSITY OF AGRICULTURAL SCIENCES

BANGALORE, GKVK

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[illegible]

Role of AI in Mitigating Biotic Stress Variability Impact on Crop Yield

[illegible]

Submitted by:

[illegible] – [illegible]

[illegible] – [illegible]

[illegible] – [illegible]

Submitted to:

Dr. Kumar [illegible]

Assoc University

Submitted on: [illegible]

UNIVERSITY OF AGRICULTURAL SCIENCES DHARWAD

(Kalli Tryanna - 2020)

AREA BACHELOR THESIS

(B. Tech. (Agri))

Role of Artificial Intelligence in Mitigating the Impact of Climate Change Variables on Land Yield of Tomato Fruit (Lycopersicum Esculentum) in Kalthy, Pri District, Kerala

(Submitted in partial fulfilment of the requirements for the degree of Bachelor of Technology (Agriculture) of the University of Agricultural Sciences, Dharwad)

[illegible]	[illegible]
[illegible]	[illegible]
[illegible]	[illegible]
[illegible]	[illegible]
[illegible]	[illegible]

[illegible]

1. Abstract

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

1.1 [illegible]

[illegible]

[illegible]

[illegible]

[illegible]

2. Objectives

[illegible]

Specific Objectives

- [illegible]
- [illegible]
- [illegible]
- [illegible]
- [illegible]

3. [illegible] Digital Signature

[illegible]

[illegible]

The Fifth Algorithm — [illegible] — [illegible]

[illegible]

The Sixth Algorithm — [illegible] — [illegible]

[illegible]

The Seventh Algorithm — [illegible] — [illegible]

[illegible]

[illegible]

1. Project Summary

1.1 Real-World Problem Context

[illegible]

[illegible]

1.1.1 The Global Economic Burden — A Silent Epidemic

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]	[illegible]
[illegible]	[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[

6. Dataset Description

6.1 Data Sources

- [illegible]
- [illegible]
- [illegible]

6.1.1 Data Quality Characteristics and Quality Assessment

[illegible]

[illegible]

[illegible]

4.1 Features and Target Variables

[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]	[illegible]	[illegible]

4.2 [illegible]

[illegible]

4.3 Discussion of Selected Variables and Their Roles

[illegible]

[illegible]

[illegible]

地区	公共汽车	汽车出行	骑自行车出行	步行出行	其他	城市公共交通
南方	47	114	298	102	9	14 (1970)(2)[illegible]
南方	48	142	489	102	9	21 (1970)(2)[illegible]
南方	7	114	122	20	9	14 (1970)(2) [illegible]
南方	52	133	97	16	4	47 (1970)(2)[illegible]
南方	10	114	273	32	9	14 (1970)(2)[illegible]
南方	7	142	489	20	4	14 (1970)(2)[illegible]
南方	14	71	124	14	3	17 (1970)(2) [illegible]
北京	11	142	102	17	6	[illegible]
南方	10	144	298	20	4	11 (1970)(2)[illegible]
南方	50	214	489	24	7	[illegible]
北京	13	142	102	30	6	[illegible]
南方	21	113	190	4	1	[illegible]

[illegible]

[illegible]

3. Methodology

3.1 Data Preprocessing

[illegible]

[illegible]

[illegible]

3.1.1 Extreme Preprocessing Discussion

[illegible]

[illegible]

[illegible]

[illegible]

[illegible] Engineering

[illegible]

[illegible]

[illegible] Engineering Principles and Practices [illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

1. [illegible]
2. [illegible]
3. [illegible]
4. [illegible]
5. [illegible]

Comparison to Algorithmic Governance

Capability	Human	Agent
[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]

Note. [illegible]

1.3.1 Hyperscalable Configurations and Cross-Platform Protocol

[illegible]

[illegible]

[illegible]

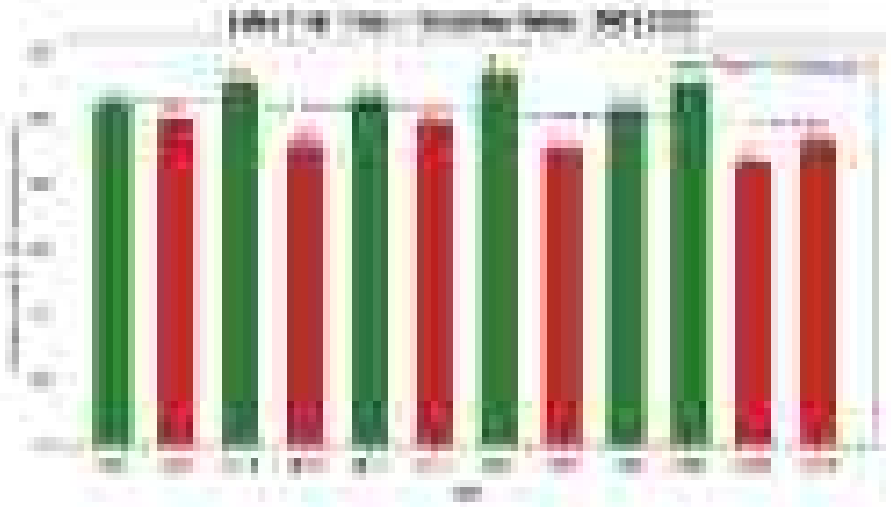
[illegible]

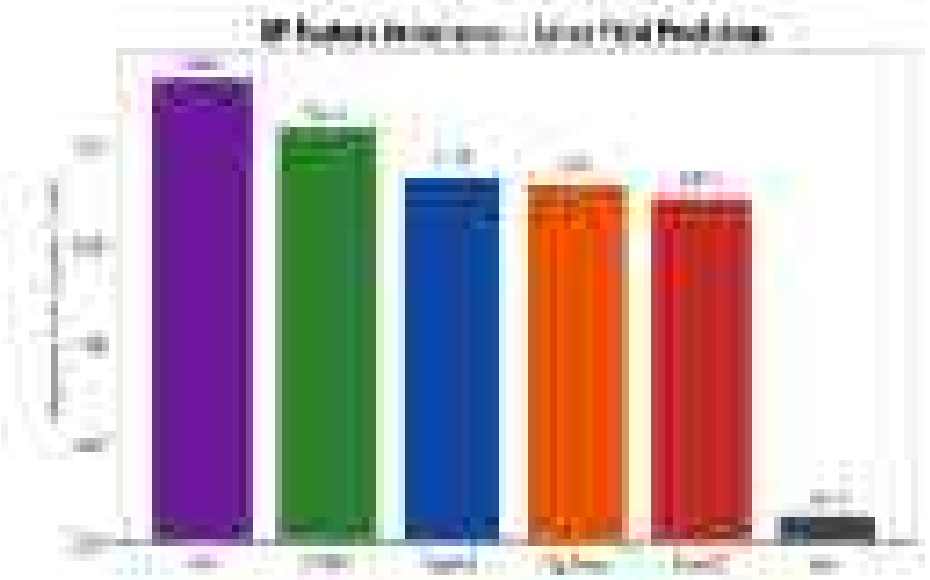
6. Google Colab – Code Screenshots

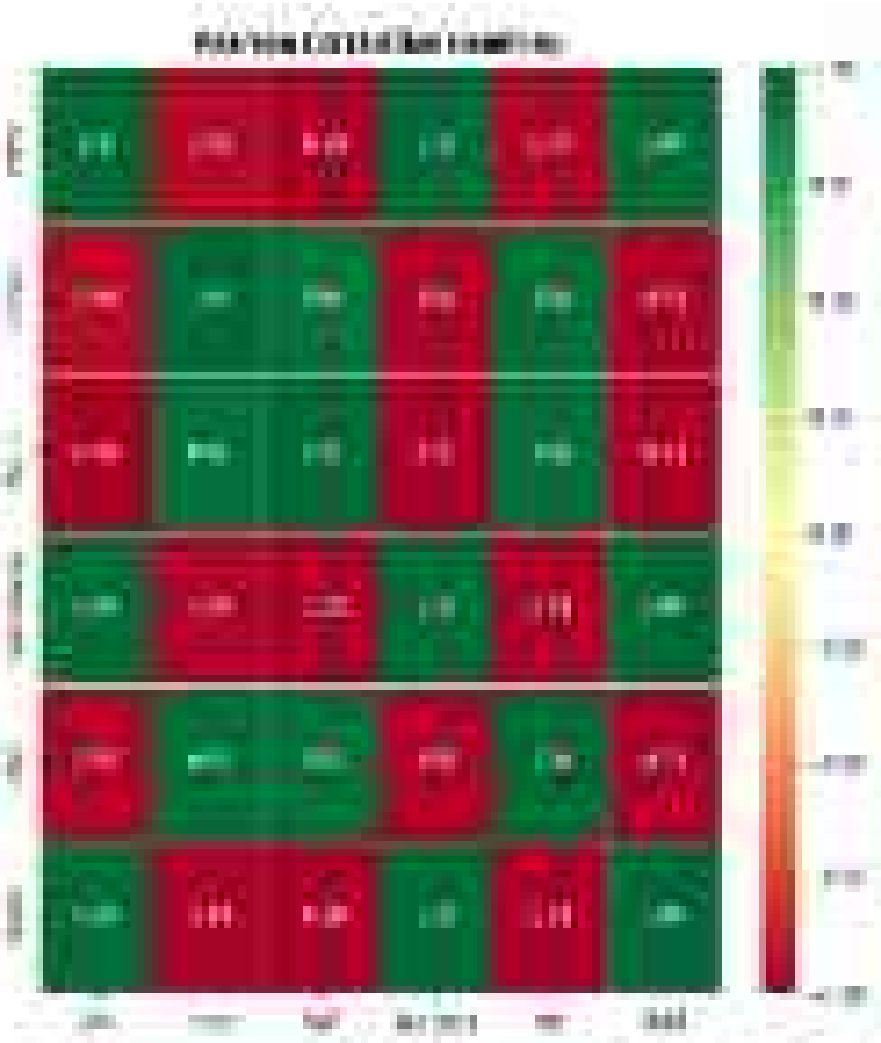
[illegible]

```
[illegible]
```

[illegible]







6. Results and Analysis

6.1 Performance Metrics

[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]

6.2 Predicted vs. Actual — Pakistan Data

[illegible]

Year	Under	Formula	Lab Free	Sales
(24)	(2.4)	2.1	2.7	Nonfiction
(47)	47	2.1	1.2	Alkylsulfonic
(4.1)	(4.1)	5.7	−1	Hydrofluoric acid + Iron(II) chloride
(26)	47	2.1	−1.7	Oxygen + Iron + Iron(II) oxide
(47)	(5.2)	2.1	2.4	Nonfiction
2.1	47	5.1	1.7	Bleaching agent
(2.1)	47	2.1	−1.2	Mg + Cl2 → (mole ratio)
(2.1)	(4.1)	2.1	−1.4	Nitric acid + Iron + Iron(II) oxide
2.22	5.1	2.1	2.1	Nonfiction
(26)	(2.4)	−2.1	−1.7	Mg + Cl2 → (mole ratio)
1.2	−2.4	4.1	2.1	Fe2O3 + Al → Al2O3 + Fe (mole ratio)
(26)	−2.4	5.1	1.2	Ammonium sulfate

1 2 3 | sales in http://ufo.com

Rules	Problem	Assessment	Investigation
1	[illegible]	[illegible]	[illegible]
2	[illegible]	[illegible]	[illegible]
3	[illegible]	[illegible]	[illegible]
4	[illegible]	[illegible]	[illegible]
5	[illegible]	[illegible]	[illegible]
6	[illegible]	[illegible]	[illegible]

[illegible]

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[illegible]

5. Discussion

5.1 [illegible]

[illegible]

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5.2.1 [illegible]

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[illegible]

1.4 [illegible]

[illegible]

[illegible]

8. Discussion

8.1 Scientific Contribution

- [illegible]
- [illegible]
- [illegible]
- [illegible]

8.2 Practical Implications

[illegible]

8.3 Limitations and Future Work

[illegible]

8.4 Concluding Remarks

[illegible]

[illegible] Pradesh (MP) [illegible] is a major [illegible] agricultural [illegible] [illegible] [illegible]

[illegible] Applications in Agriculture

[illegible]

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11. References

[1] [illegible]

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